

Copper Electroplating

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Students visibly transform a metal object

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Preparation
3. Materials checklist
4. Experiment procedure
5. Shared safety notes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

Session 2 — Copper Electroplating

Hook: Students visibly transform a metal object.

Learning outcomes

- Explain electrolysis and distinguish it from a galvanic cell
- Identify cathode and anode in an **electrolytic** cell
- Describe how current controls deposition rate
- Explain why surface preparation and current density affect deposit quality

Session plan (90 min)

Time	Activity	Reference
0–10 min	Inspect overnight copper solution; discuss color and dissolution evidence	lecture.md — Hook
10–25 min	Concept: ion deposition at cathode, anode dissolution	lecture.md
25–35 min	Clean cathode object (sand, rinse, no fingerprints)	experiment.md — Prep
35–50 min	Set up plating cell	experiment.md — Setup
50–70 min	Plate 2–10 min; observe every 1–2 min	experiment.md — Plating
70–80 min	Compare variables	experiment.md — Variables
80–90 min	Discussion: surface prep and current density	lecture.md — Wrap-up

Key reactions

Cathode (–): $\text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-} \rightarrow \text{Cu}(\text{s})$

Anode (+): $\text{Cu}(\text{s}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2\text{e}^{-}$

Files

- lecture.md — Concept block
- experiment.md — Plating procedure
- materials.md — Checklist
- preparation.md — Solution prep and backup plan

Status

- [] Copper solution from Session 1 inspected
- [] Backup CuSO_4 solution prepared
- [] Plating procedure pilot-tested
- [] Lecture slides/diagrams ready

Session 2 — Preparation

Depends on Session 1

Copper solution should be started at end of Session 1. See ../01-fruit-battery/preparation.md.

Before Session 2 (instructor)

- ☐ Inspect all overnight jars — note color and clarity
- ☐ Prepare **backup 0.5 M CuSO₄** (e.g. 62 g CuSO₄·5H₂O per 500 mL tap or distilled water); label concentration
- ☐ Pilot plate one nail: record time, current, appearance
- ☐ Pre-sand a few demo objects
- ☐ Source resistors if using current limiting

Homemade solution quality guide

Appearance	Action
Pale blue/green, clear	Likely usable
Very weak color	Use backup or extend prep time
No color change	Switch to CuSO ₄ backup

Day-of setup

- ☐ Goggles and gloves at each station
- ☐ One 9 V battery per group (fresh if possible)
- ☐ Waste bucket labeled "copper waste — do not drain"
- ☐ Demo cell ready for hook (jar + electrodes + meter)

Backup plan

If homemade solution fails for multiple groups:

1. Switch to dilute CuSO₄ backup
2. Explain that industrial plating uses optimized baths — ours is "kitchen chemistry"

Open questions / notes

- Backup CuSO₄ concentration: 0.5 M (or note if different):
- Resistor value used in pilot:
- Best demo object (nail vs. key):

Session 2 — Materials

Per group

- ☐ Copper solution — homemade from Session 1 or **0.5 M CuSO₄** backup (50–100 mL)
- ☐ Copper gasket, strip, or wire (anode)
- ☐ Steel nail, screw, coin, washer, or key (cathode)
- ☐ 9 V battery or low-voltage DC supply
- ☐ Alligator clips
- ☐ Multimeter
- ☐ Optional: 100–500 Ω resistor (target ~10–40 mA)
- ☐ Sandpaper or steel wool
- ☐ Jam jar or beaker (~100 mL)
- ☐ Gloves and goggles

Instructor / backup — 0.5 M CuSO₄

Batch	CuSO ₄ ·5H ₂ O	Water
200 mL	25 g	to 200 mL
500 mL	62 g	to 500 mL
1 L	125 g	to 1 L

- ☐ Extra copper anode material
- ☐ Waste container for copper-bearing liquids
- ☐ Paper towels for rinsing/drying

From Session 1 carryover

- ☐ Overnight vinegar/salt/copper jars — inspect before class

Pilot notes

Parameter	Target from pilot
Solution	0.5 M CuSO ₄ or homemade
Solution volume	~50–100 mL
Typical current (with resistor)	___ mA
Good plating time	___ min

Session 2 — Experiment: Copper Electroplating

Instructors: polarity checks, deposit quality, and solution notes — instructor.md.

Copper solution — what to use

You need a blue **copper(II)** solution. Use either the overnight homemade bath from Session 1 or the instructor CuSO_4 backup.

Option A — Homemade (from Session 1)

Ingredient	Amount per jar
White vinegar	~100 mL
Table salt	1–2 tsp (~5–10 g)
Clean copper metal	Large surface area (wire coil, gasket, or strips)

- Started at end of Session 1; should look pale blue/green after overnight.
- If almost colorless → use Option B.

Option B — Copper sulfate backup (instructor prepares)

Target concentration: 0.5 M CuSO_4 (recommended classroom strength).

Batch size	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (blue crystals)	Water
200 mL (about 2 groups)	25 g	Fill to 200 mL
500 mL	62 g	Fill to 500 mL
1.0 L	125 g	Fill to 1.0 L

Water: tap water is fine. Distilled is slightly better if your tap water is very hard (bath may look milky). Save distilled water for Session 3 silver nitrate.

How to mix:

1. Weigh $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ (copper sulfate pentahydrate — the common blue lab salt).
2. Dissolve in $\sim 3/4$ of the water; stir until clear blue.
3. Add water to the final volume; label "**0.5 M CuSO_4 — Session 2 — do not drink**".
4. Store covered; goggles/gloves when pouring.

Acceptable range if you must improvise: 0.25–1.0 M. Below ~ 0.25 M plating is slow; much above 1 M can give rough deposits and waste chemical.

Do not add strong acid (industrial baths sometimes use H_2SO_4 — skip that for this age group).

Volume per group today

50–100 mL of copper solution in a jam jar or small beaker (enough to immerse both electrodes).

Prep — Clean the cathode (25–35 min)

Object options

Steel nail, screw, coin, washer, or key (one “good” object + optional unplated control of the same type).

Procedure

1. Sand or steel-wool the surface until **bright metal** shows (no rust, paint, or dull oxide).
2. Rinse under water; pat dry with a paper towel.
3. **Do not touch** the area that will be plated — hold by the head, thread, or edge only. Finger oil ruins adhesion.
4. Optional: photograph or weigh before plating for comparison.

Quality checklist

- ☐ No visible rust or grease
 - ☐ Uniform bright sanded surface
 - ☐ Rinsed and dry
 - ☐ Placed on a clean paper towel until use
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Setup — Plating cell (35–50 min)

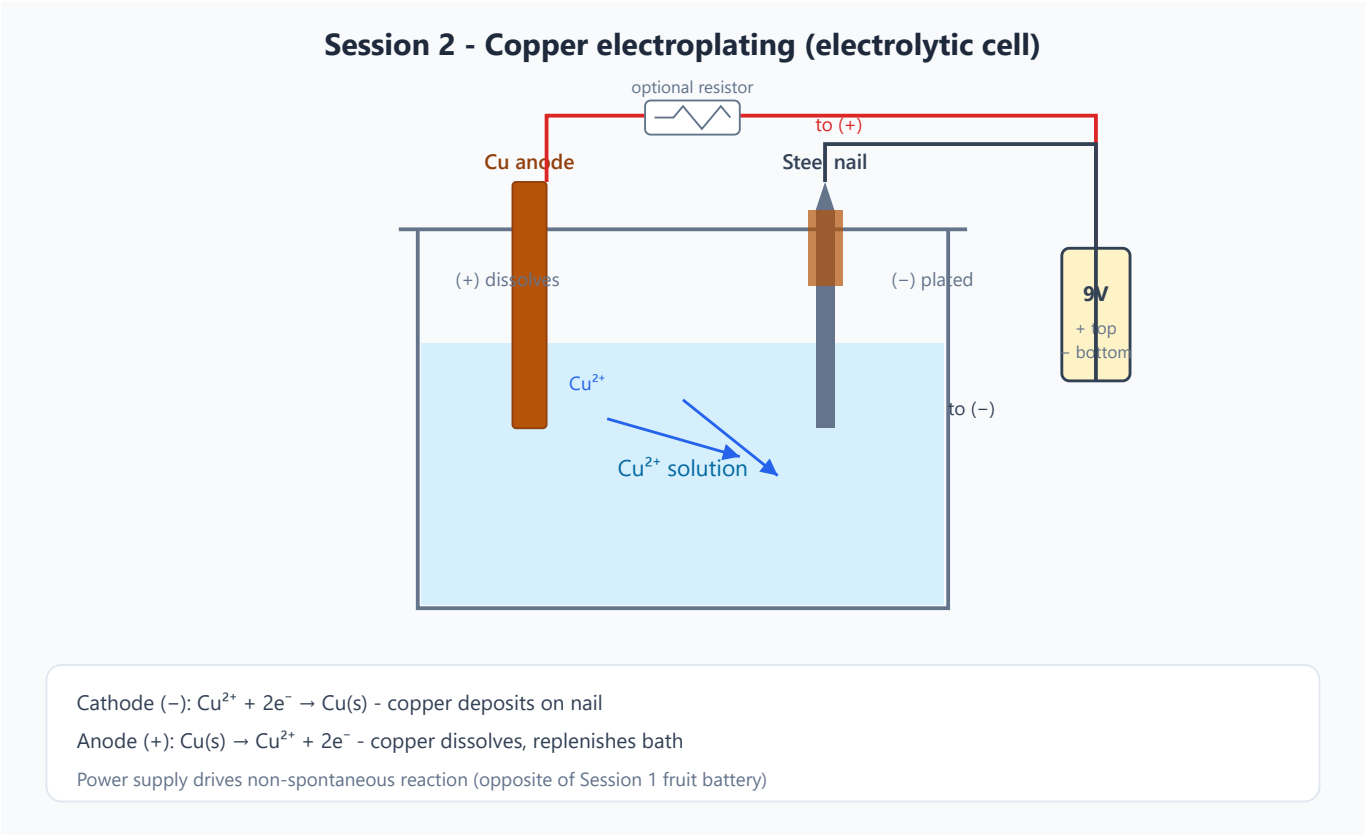


Figure 1 — Electrolytic cell: Cu anode (+) dissolves; Cu^{2+} deposits on the nail cathode (-). Optional resistor limits current.

Materials at the station

- 50–100 mL copper solution (Option A or B)
- Copper strip, gasket, or thick wire (**anode**)
- Cleaned object (**cathode**)
- 9 V battery (fresh)
- Alligator clips (at least 2)
- Optional: 100–500 Ω resistor, multimeter on DC mA

Wiring — follow exactly

Part	Connects to
Object to plate (nail, etc.)	Battery negative (-) / black
Copper anode	Battery positive (+) / red
Optional resistor	In series on either lead (limits current)
Optional ammeter	In series on DC mA range

Memory aid: *The object that should gain copper goes to the minus sign.*

Procedure

1. Pour **50–100 mL** copper solution into the jar.
2. Clip the **copper anode** to battery **(+)**; clip the **object** to battery **(–)**.
3. If using a resistor: put it in series so current must pass through it.
4. If using a multimeter for current: set to DC mA; wire it **in series** (not across the battery like a voltmeter).
5. Lower both electrodes into the solution so the plating area is submerged.
6. Keep electrodes **2–4 cm apart**; they must **not touch** (touching = short circuit).
7. Confirm bubbles or a slow color change starting within ~30–60 s. Read initial current if possible (**target ~10–40 mA** with a resistor; without a resistor current may be higher).

Pre-plating record

Item	Value
Solution (homemade / 0.5 M CuSO ₄)	
Approximate concentration if known	
Solution volume (mL)	
Solution color	
Anode material	
Cathode object	
Electrode spacing (cm)	
Initial current (mA)	
Resistor used? (Y/N, Ω)	

Plating run (50–70 min)

1. Start the timer when the circuit is connected and current is flowing.
2. Every **1–2 minutes**, look at the object: color, shine, roughness, bubbles.
3. Typical good run: **3–8 minutes**. Stop sooner if the deposit turns black or powdery.
4. Disconnect the battery **before** removing the object.
5. Rinse gently under water; pat dry. Do not scrape the new copper.
6. Compare to an unplated control of the same object type.

Observation log

Time (min)	Appearance	Current (mA)	Notes
0			
2			
4			

Time (min)	Appearance	Current (mA)	Notes
6			
8			
Final			

What “good” looks like

Appearance	Meaning	Action
Bright salmon / copper color	Good plating	Continue or stop at planned time
Dark brown / black powder	Current too high	Stop; add resistor; re-sand; restart shorter
Patchy	Dirty surface or poor immersion	Re-clean; ensure full immersion
Heavy bubbles on object	Competing H ₂ evolution	Lower current / increase spacing

Variable comparisons (70–80 min)

Groups test **one** variable (share results):

Variable	Options	What to compare
Plating time	2 vs. 8 min	Thickness, color
Current	With vs. without 220–470 Ω resistor	Smoothness
Surface prep	Clean vs. finger-greased	Adhesion, coverage
Solution	Homemade vs. 0.5 M CuSO ₄	Speed, color quality
Distance	~1 cm vs. ~4 cm electrode gap	Uniformity, current

Results table (class compilation)

Group	Variable	Result	Best practice conclusion

Troubleshooting

Problem	Cause	Fix
No plating	Reversed polarity	Object must be on (–)
No plating	Solution too weak	Switch to 0.5 M CuSO ₄ backup
Black/powdery Cu	Current too high	Add 100–500 Ω resistor; increase distance
Patchy deposit	Dirty / oily surface	Re-sand, rinse, no fingerprints
Anode not dissolving	Wrong anode (not copper)	Use copper strip/wire

Problem	Cause	Fix
Battery gets hot	Short circuit	Separate electrodes; check clips
Very slow plating	Low $[\text{Cu}^{2+}]$ or bad contacts	Use backup bath; clean clips

Safety during experiment

- Goggles on when working with solutions; gloves recommended
 - Do not ingest solutions; no mouth pipetting
 - Avoid short-circuiting the 9 V battery
 - Collect all copper-bearing liquid in the labeled waste container — **not** down the drain
 - Wash hands after the lab
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Experiment status

- ☐ 0.5 M CuSO_4 backup mixed and labeled (or homemade jars verified)
- ☐ Optimal plating time determined in pilot (record: ___ min)
- ☐ Resistor value chosen for ~10–40 mA (record: ___ Ω)
- ☐ Before/after photos for lecture use
- ☐ Variable comparison sheet printed for groups

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M AgNO₃** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution	No saltwater; no flames near setup; instructor-only pop test
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H₂/O₂ container